

## BRAF mutant non-small cell lung cancer and treatment with BRAF inhibitors.

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Inhibitors targeting active protein kinases, such as EGFR or ALK, have demonstrated significant efficacy in the treatment of lung cancer. Activating mutations in the MAPK pathway, which includes the enzymes RAS, RAF, MEK, and ERK, result in constitutive signalling, leading to oncogenic cell proliferation and escape from apoptosis; therefore this pathway is a focus of crucial interest for the development of cancer drugs. In melanoma, the most commonly mutated gene is BRAF, with mutations usually occurring in about 50% of all tumours. The BRAF Val600Glu (V600E) mutation constitutes more than 90% of mutations in melanoma. V600E BRAF mutation shows a great dependency on MEK activity, and offers a rational therapeutic strategy for this genetically defined tumour subtype. The use of vemurafenib and dabrafenib, agents that block MAPK signaling in patients with melanoma and the BRAF V600E mutation, has been associated with prolonged survival and progression-free survival. The frequency of V600E BRAF mutation in lung adenocarcinoma is 1.5% to 2.8%. Treatment of V600E BRAF-mutant lung adenocarcinomas with dabrafenib is under evaluation in a phase 2 trial, and could represent another milestone in individualized therapy for lung cancer patients. The next step will be a combination therapy of BRAF inhibitor dabrafenib and MEK inhibitor trametinib.

## Modelling vemurafenib resistance in melanoma reveals a strategy to forestall drug resistance.

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Mutational activation of BRAF is the most prevalent genetic alteration in human melanoma, with  $\geq 50\%$  of tumours expressing the BRAF(V600E) oncoprotein. Moreover, the marked tumour regression and improved survival of late-stage BRAF-mutated melanoma patients in response to treatment with vemurafenib demonstrates the essential role of oncogenic BRAF in melanoma maintenance. However, as most patients relapse with lethal drug-resistant disease, understanding and preventing mechanism(s) of resistance is critical to providing improved therapy. Here we investigate the cause and consequences of vemurafenib resistance using two independently derived primary human melanoma xenograft models in which drug resistance is selected by continuous vemurafenib administration. In one of these models, resistant tumours show continued dependency on BRAF(V600E)→MEK→ERK signalling owing to elevated BRAF(V600E) expression. Most importantly, we demonstrate that vemurafenib-resistant melanomas become drug dependent for their continued proliferation, such that cessation of drug administration leads to regression of established drug-resistant tumours. We further demonstrate that a discontinuous dosing strategy, which exploits the fitness disadvantage displayed by drug-resistant cells in the absence of the drug, forestalls the onset of lethal drug-resistant disease. These data highlight the concept that drug-resistant cells may also display drug dependency, such that altered dosing may prevent the emergence of lethal drug resistance. Such observations may contribute to sustaining the durability of the vemurafenib response with the ultimate goal of curative therapy for the subset of melanoma patients with BRAF mutations.

## Targeting BRAF in advanced melanoma: a first step toward manageable disease.

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Melanoma is the deadliest form of skin cancer and its incidence has been increasing worldwide. The disease manifests itself as clinically and genetically distinct subgroups, indicating the need for patient-specific diagnostic and treatment tools. The discovery of activating mutations (V600E) in the BRAF kinase in approximately 50% of patients spurred the development of compounds to inhibit aberrant BRAF activity, and the first drug candidate to show promising clinical activity is PLX4032 (also known as RG7204). Most recent clinical data from a phase II trial indicate that PLX4032 causes tumor regression and stabilized disease in >50% of advanced melanoma patients harboring BRAF V600E tumors. These data validate the effectiveness of oncogene-targeted therapy against advanced melanoma and offer hope that the disease can be overcome. However, as melanoma is dynamic and heterogeneous, careful treatment strategies and combination therapies are warranted to obtain long-term clinical effects.

## BRAF Mutant Melanoma Adjusts to BRAF/MEK Inhibitors via Dependence on Increased Antioxidant SOD2 and Increased Reactive Oxygen Species Levels.

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B-Rapidly Accelerated Fibrosarcoma (BRAF) mutations are found in about 50% of melanoma patients. Treatment with Food and Drug Administration (FDA)-approved BRAF and MAP/ERK kinase (MEK) inhibitors has improved progression free and overall survival of patients with BRAF mutant melanoma. However, all responders develop resistance typically within 1 year of treatment with these inhibitors. Evidence indicates that reactive oxygen species (ROS) levels are elevated after BRAF pathway inhibition treatment. We aim to decipher the role of mitochondrial antioxidant proteins relative to ROS levels and BRAF pathway inhibitor resistance. We observed BRAF mutant melanoma cells treated with the combination of a MEK inhibitor (trametinib) and a BRAF inhibitor (dabrafenib), exhibited elevated ROS levels, both in in vitro and in vivo melanoma models. We next generated trametinib- and dabrafenib-resistant (TDR) cells and found increased ROS levels after acquisition of resistance. An immunofluorescence experiment showed an increase of DNA damage in TDR cell lines. Furthermore, we observed that TDR cells increased superoxide dismutase 2 (SOD2), an antioxidant, at both mRNA and protein levels, with the upregulation of the transcription factor Nuclear Factor (NF)- $\kappa$ B. Knockdown of SOD2 significantly reduced the growth of BRAF pathway inhibitor-resistant cells. In addition, the results indicate that TDR cells can be re-sensitized to BRAF pathway inhibitors by the ROS scavenger, N-Acetyl Cysteine (NAC). Overall, these data indicate that BRAF pathway inhibitor-resistant cells can compensate for elevated ROS via increased expression of the antioxidant SOD2.

## Targeting the p300/NONO axis sensitizes melanoma cells to BRAF inhibitors.

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BRAF inhibitors (BRAFi) that target BRAF V600E kinase, a driver mutation found in 50% of melanomas, show a significant antitumor response, but the common emergence of acquired resistance remains a challenge. Abnormal expression of RAF isoforms CRAF and ARAF reactivates pERK1/2, which plays crucial roles in the acquisition of resistance of melanoma cells. However, the mechanisms of dysregulation of RAF isoforms in resistant melanoma cells remain unknown. Here, we identified NONO interacted with and stabilized both CRAF and ARAF in melanoma cells, and that NONO was acetylated at 198K by p300 acetyltransferase, which stabilized NONO via antagonizing its ubiquitination/degradation mediated by

RNF8. The upregulation of both p300 and NONO promoted the rebound of pERK1/2 and the subsequent resistance of melanoma cells to BRAFi, and the activation of ERK1/2 in turn induced p300 to form a positive feedback loop in resistant melanoma cells. There was a positive correlation between p300 and NONO in resistant melanoma cells and clinical samples, and p300 inhibitor C646 overcame the resistance of resistant melanoma cells to BRAF inhibitors in vitro and in vivo. Our findings reveal that targeting the positive feedback loop of p300-NONO-CRAF/ARAF-pERK1/2 may be excellent strategies to overcome the resistance of BRAF inhibitors for melanoma patients.

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## Novel and Potent Small Molecules against Melanoma Harboring BRAF Class I/II/III Mutants for Overcoming Drug Resistance.

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Melanoma accounts for the majority of skin cancer deaths. About 50% of all melanomas are associated with BRAF mutations. BRAF mutations are classified into three classes with regard to dependency on RAF dimerization and RAS signaling. The most frequently occurring class I BRAF V600 mutations are sensitive to vemurafenib whereas class II and class III mutants, non-V600 BRAF mutants are resistant to vemurafenib. Herein we report six pyrimido[4,5-d]pyrimidin-2-one derivatives possessing highly potent anti-proliferative activities on melanoma cells harboring BRAF class I/II/III mutants. Novel and most potent derivative, SIJ1777, possesses not only two-digit nanomolar potency but also 2 to 14-fold enhanced anti-proliferative activities compared with reference compound, GNF-7 against melanoma cells (SK-MEL-2, SK-MEL-28, A375, WM3670, WM3629). Moreover, SIJ1777 substantially inhibits the activation of MEK, ERK, and AKT and remarkably induces apoptosis and significantly blocks migration, invasion, and anchorage-independent growth of melanoma cells harboring BRAF class I/II/III mutations while both vemurafenib and PLX8394 have little to no effects on melanoma cells expressing BRAF class II/III mutations. Taken together, our six GNF-7 derivatives exhibit highly potent activities against melanoma cells harboring class I/II/III BRAF mutations compared with vemurafenib as well as PLX8394.

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## The significance of BRAF V600E mutation status discordance between primary cutaneous melanoma and brain metastases: The implications for BRAF inhibitor therapy.

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To compare BRAF V600E status of primary melanoma and brain metastases to assess for discordance by cross-sectional study, and to evaluate clinical implications on BRAF inhibitor therapy. Brain metastases are common in patients with advanced melanoma. Between 40% and 60% of melanomas demonstrate BRAF mutations, BRAF V600E being most common. Selective BRAF inhibitor therapy has shown improvement in outcome in patients with melanoma. It has been demonstrated that not all metastatic lesions carry the same BRAF mutation status as the primary, but the frequency in which discordance occurs remains unclear. Establishing this may have implications in the use of BRAF inhibitors in patients with melanoma brain metastases. Patients who underwent metastectomy for melanoma brain metastases were identified using our local histopathology database. A review of histology of the primary lesion and the metastasis was performed for each patient, assessing for BRAF mutation status discordance. Forty-two patients who underwent a brain metastectomy following excision of a melanoma

primary were identified over a 7-year period. Median survival was 9 months. The median Breslow thickness for the primary lesion was 3.4mm. Six patients (14%) had discrepancy between the BRAF status of a melanoma primary and metastatic lesion. Of these 6 patients, 3 had a BRAF mutation positive primary with a BRAF mutation negative metastatic lesion, while the other 3 had a BRAF mutation negative primary with BRAF mutation positive metastasis. There is an important discordance rate in the BRAF mutation status of melanoma primaries versus brain metastases.

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## BRAF mutations in metastatic malignant melanoma: comparison of molecular analysis and immunohistochemical expression.

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Melanoma is a complex genetic disease, and multiple genetic alterations have been reported to play a role during disease progression. The dysregulation of BRAF signaling has been shown to affect many molecules that promote the continual progression of melanoma. Oncogenic BRAF expression plays a vital role in promoting cell invasion and metastasis in melanoma. It is also associated with poor prognosis in metastatic melanoma. About 40% to 60% of cutaneous melanomas have BRAF mutations, and 90% of the mutations involve a specific substitution at codon 600 (BRAF V600E). In this study, we compared BRAF (V600E) mutation detection by molecular analysis with BRAF expression by immunohistochemical (IHC) analysis using 2 different antibodies. A total of 25 metastatic malignant melanomas were included in this study. 10 of the 25 (40%) cases were positive by molecular analysis using the COBAS 4800 BRAF V600 Real-time PCR assay, and 18 of the 25 (72%) cases were positive by IHC analysis with 2 different antibodies. All positive cases by molecular analysis were positive by both IHC stains (100%). No false negatives were obtained with either IHC stain. Eight of the 25 (32%) were false-positive by both IHC stains. This study demonstrates that IHC analysis is a very sensitive test for evaluation of BRAF mutations in metastatic malignant melanoma and may be useful as an initial screening test.

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## Oxidative Stress-Related Mechanisms in Melanoma and in the Acquired Resistance to Targeted Therapies.

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Melanoma is a highly aggressive cancer with the poorest prognosis, representing the deadliest form of skin cancer. Activating mutations in BRAF are the most frequent genetic alterations, present in approximately 50% of all melanoma cases. The use of specific inhibitors towards mutant BRAF variants and MEK, a downstream signaling target of BRAF in the MAPK pathway, has significantly improved progression-free and overall survival in advanced melanoma patients carrying BRAF mutations. Nevertheless, despite these improvements, resistance still develops within the first year of therapy in around 50% of patients, which is a significant problem in managing BRAF-mutated advanced melanoma. Understanding these mechanisms is one of the mainstays of the research on BRAFi/MEKi acquired resistance. Both genetic and epigenetic mechanisms have been described. Moreover, in recent years, oxidative stress has emerged as another major force involved in all the phases of melanoma development, from initiation to progression until the onsets of the metastatic phenotype and chemoresistance, and has thus become a target for therapy. In the present review, we discuss the current knowledge on oxidative stress and its signaling in melanoma, as well as the oxidative

stress-related mechanisms in the acquired resistance to targeted therapies.

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## Cobimetinib in malignant melanoma: how to MEK an impact on long-term survival.

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Approximately 50% of cutaneous melanomas harbor activating mutations of the BRAF-oncogene, making BRAF inhibitors (BRAFi) the standard treatment for this disease. However, disease responses are limited in duration mainly due to acquired resistance. Dual MAPK pathway inhibition with addition of a MEK inhibitor (MEKi) to a BRAFi improved the efficacy and tolerability compared with BRAFi alone. Cobimetinib (Cotellic®) is an orally bioavailable, potent and selective MEKi, which significantly improved response rates when combined with BRAFi vemurafenib (median overall survival: 22.3 months). The toxicity profile of cobimetinib is manageable and treatment discontinuation due to adverse events is uncommon. Present efforts are addressed to overcome resistance and improve long-term outcomes: based on the evidence of the immunomodulatory properties of BRAFi and MEKi, current clinical trials of combined targeted and immunotherapy are investigating the role of cobimetinib in the context of combination or as sequential treatments.

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